

A Comparative Study of Transfer Learning Strategies for Low-Resource Sign Language Recognition: From ASL to PSL

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https://github.com/sameeha236/DL_Project

Abstract

Pakistani Sign Language (PSL) serves millions of deaf and hard-of-hearing individuals in Pakistan, yet the largest public dataset (WLPSL) contains only 248 videos across 31 word classes—far too little to train deep models from scratch. American Sign Language (ASL), by contrast, benefits from large-scale resources such as WLASL [1]. We investigate whether knowledge learned from ASL can be transferred to PSL, and which combination of architecture and transfer strategy is most effective under severe data constraints.

We implement a unified four-stage transfer learning pipeline—zero-shot evaluation, linear probing, partial fine-tuning, and full fine-tuning—and apply it across three spatiotemporal architectures: RGB video (I3D), skeleton graphs (Pose-TGCN), and sequence attention (Pose Transformer). Source-domain models are pretrained on WLASL; target-domain evaluation uses WLPSL (PSL) and, for methodology validation at scale, LSFB-ISOL French Sign Language poses (top-100 glosses).

On FSL, a Pose Transformer pretrained on WLASL pose features achieves 59.30% top-1 accuracy (full fine-tuning), with linear probing reaching 58.27%. On PSL, zero-shot accuracy is near chance ($\sim 4\%$) across all architectures. The best PSL result is 30.00% top-1 using Pose Transformer with full fine-tuning and pose augmentation, outperforming ASL-pretrained Pose-TGCN (26.00%, partial FT) and I3D (4.00%). Our benchmark matrix and per-class analyses provide the first systematic comparison of cross-lingual transfer strategies for low-resource sign language recognition.

1 Introduction

Sign language recognition (SLR) has progressed rapidly for American Sign Language thanks to large video datasets and deep spatiotemporal models [1]. Pakistani Sign Language lacks comparable resources. WLPSL provides only 248 word-level videos spanning 31 glosses, which is insufficient for training generalizable systems without overfitting.

We treat PSL recognition as a domain adaptation problem. ASL and PSL share physical signing properties—handshape, motion, and spatial configuration—so representations learned on ASL may partially transfer. Our central question is which

transfer strategy and which architecture yield the best PSL accuracy when only 248 training videos are available.

1.1 Problem and Motivation

An estimated 10 million deaf and hard-of-hearing people in Pakistan depend on PSL. Automated word-level recognition could support education, accessibility tools, and dictionary search—but only if methods work with available data. Training exclusively on WLPSL causes overfitting; transfer learning from ASL is a natural alternative that has not been systematically benchmarked for PSL.

1.2 Overview of Results

Experiment	Dataset	Best Top-1
ASL source validation (I3D)	WLASL-100 (158 vids)	31.65%
ASL→FSL benchmark	LSFB-ISOL top-100	59.30%
ASL→PSL Pose-TGCN	WLPSL (198/50)	26.00%
ASL→PSL Transformer	WLPSL (198/50)	30.00%
ASL→PSL I3D	WLPSL (198/50)	4.00%

Table 1: Summary of key experimental results.

Code: https://github.com/sameeha236/DL_Project

1.3 Contributions

1. A reusable four-strategy transfer learning benchmark across video, graph, and transformer architectures.
2. The first systematic ASL→PSL transfer comparison on WLPSL.
3. Scale validation via ASL→FSL on 120,739 pose instances.
4. Quantitative evidence that linear probing suffices at FSL scale but partial/full FT with augmentation is required for PSL.
5. Per-class and confusion-matrix analyses for transferable vs. data-hungry PSL signs.

2 Related Work

Li *et al.* [1] introduced WLASL, the largest public word-level ASL dataset (21,083 videos, 2,000 glosses), and established both appearance-based (I3D) and pose-based (Pose-TGCN) deep learning as strong baselines. Their WLASL-100 subset and released pretrained checkpoints form our ASL source domain.

Duarte *et al.* [2] introduced How2Sign, a large-scale multimodal dataset for continuous American Sign Language, reinforcing the importance of pretrained video backbones and large-scale ASL resources. Together with WLASL, this motivates transfer learning from data-rich ASL to low-resource target languages.

For PSL, the WLPSL dataset (248 videos, 31 classes) remains the largest public word-level resource but is insufficient alone for generalizable deep learning. No prior work has systematically benchmarked ASL-to-PSL transfer strategies—the gap our study addresses.

3 Data

3.1 Source Domain — ASL (WLASL)

We use WLASL v0.3 [1]: WLASL-100 for video and pose experiments, official I3D ASL-100 checkpoint (65.89% reported top-1), and WLASL pose pretraining (3,607 glosses, 50.72% validation accuracy).

3.2 Intermediate Target — FSL (LSFB-ISOL)

LSFB-ISOL MediaPipe poses (~ 1 GB) with top-100 glosses: 68,262 training and 52,477 test sequences (64 frames, 75 landmarks $\times$ 3 coordinates).

3.3 Primary Target — PSL (WLPSL)

248 videos, 31 glosses; 198/50 train-test split (seed 42). Pose: 55 nodes, 64 frames. Video (I3D): 224×224 RGB, 64 frames. Training augmentation: scale, translation, noise (pose); horizontal flip (video). Random baseline: 3.23%.

4 Methods

4.1 Transfer Learning Strategies

Zero-shot: no target training. **Linear probe:** head only. **Partial FT:** last layers + head. **Full FT:** all parameters.

4.2 Architectures

I3D: RGB clips, WLASL-100 checkpoint, 31-class PSL head. **Pose-TGCN:** 55-node graph, ASL-100 ckpt.pth, 6 compatible layers transferred. **Pose Transformer:** 64-frame sequences; WLASL pose pretrain (FSL), 110-dim/frame input on PSL; $d_{\text{model}} = 256$, 4 layers, 8 heads, 3.37M parameters.

5 Experiments and Results

5.1 ASL Source Validation

I3D is validated on 158 available WLASL-100 test videos using the official checkpoint chain (ImageNet $\rightarrow$ ASL-100).

Metric	Our Run	Published [1]
Top-1	31.65% (50/158)	65.89%
Top-5	60.76% (96/158)	84.11%
Top-10	72.78% (115/158)	89.92%

Table 2: ASL I3D validation on WLASL-100.

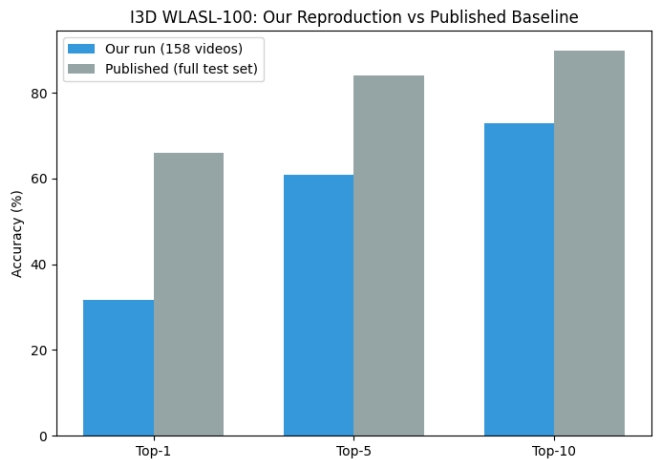


Figure 1: I3D WLASL-100 reproduction vs. published baseline.

5.2 ASL $\rightarrow$ FSL Transfer Benchmark

Pose Transformer pretrained on WLASL poses; LSFB-ISOL top-100. Random baseline: 1.0%.

Strategy	Test Top-1	Notes
Zero-shot	0.71%	Below random
Linear probe	58.27%	0.8% params trainable
Partial FT	59.23%	Last 2 blocks + head
Full FT	59.30%	Best

Table 3: FSL transfer benchmark results.

5.3 ASL $\rightarrow$ PSL Transfer Benchmark

All architectures achieve $\sim 4\%$ zero-shot (near 3.23% random), confirming PSL-specific adaptation is required.

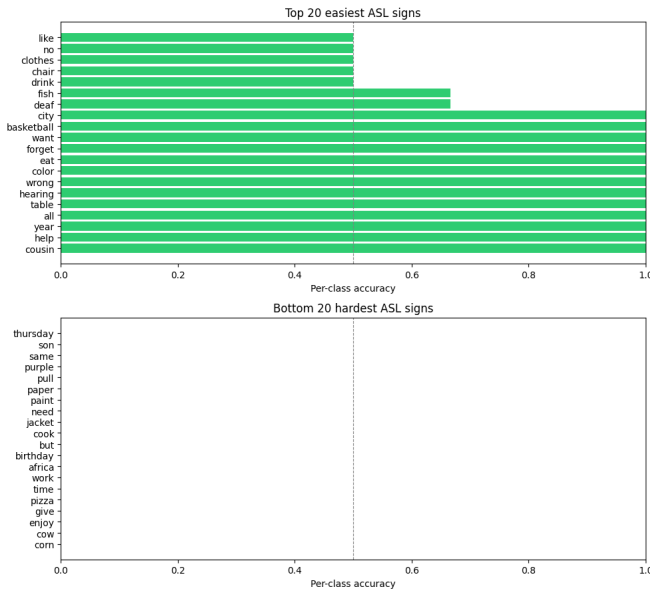


Figure 2: Per-class I3D accuracy on WLASL-100.

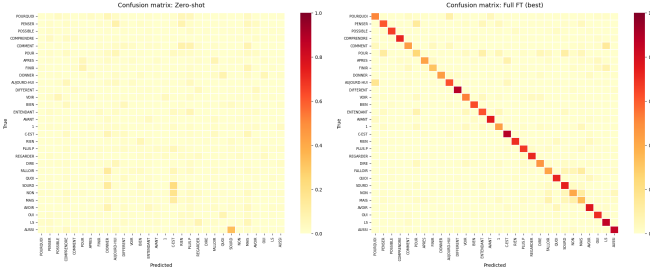


Figure 3: FSL confusion matrices across strategies.

5.4 Master Result Matrix

5.5 Key Findings

1. Zero-shot PSL ($\sim 4\%$) is near chance—adaptation is essential.
2. On FSL, linear probing captures 98% of full-FT gain; on PSL, partial/full FT with augmentation is needed.
3. Pose Transformer (30.00%) outperforms Pose-TGCN (26.00%) and I3D (4.00%) on WLPSL.
4. Pose augmentation is critical; without it, TGCN full FT underperforms linear probing.
5. ASL I3D validation (31.65% top-1) establishes a verified source pipeline before transfer.

6 Conclusion

We present the first systematic benchmark of transfer learning strategies for cross-lingual sign language recognition focused on low-resource PSL. Our four-strategy framework spans I3D, Pose-TGCN, and Pose Transformer models. FSL validation (59.30% best) confirms methodology at scale; on PSL, Pose Transformer with full fine-tuning achieves 30.00% top-1—our recommended configuration for WLPSL. This work provides

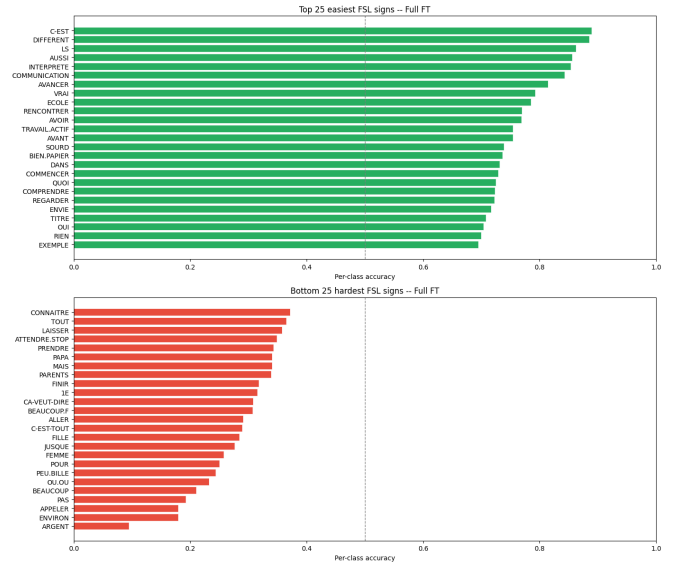


Figure 4: FSL per-class accuracy.

Arch.	Zero	Linear	Part.	Full	Best
Pose-TGCN	4.00	22.00	26.00	24.00	Part.
Transformer	4.00	6.00	26.00	30.00	Full
I3D	4.00	0.00	2.00	4.00	Zero/Full

Table 4: PSL top-1 accuracy (%) on WLPSL test split.

a reusable methodology applicable to any low-resource sign language beyond PSL.

References

- [1] D. Li, C. Rodriguez, X. Yu, and H. Li, “Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2020. arXiv:1910.11006. <https://arxiv.org/abs/1910.11006>
- [2] A. Duarte, Y. Palaskar, L. Vedula, S. Ghosh, J. Pino, and A. Metze, “How2Sign: A Large-scale Multimodal Dataset for Continuous American Sign Language,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

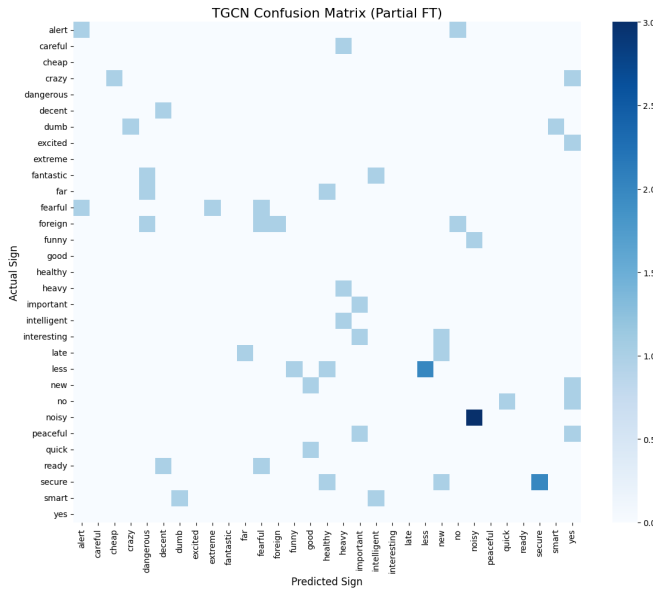


Figure 5: PSL Pose-TGCN confusion matrix (26%).

Architecture	Zero	Linear	Full	Part.
I3D (video)	4.00	0.00	4.00	2.00
Pose-TGCN (graph)	4.00	22.00	24.00	26.00
Pose Transformer (seq.)	4.00	6.00	30.00	26.00

Table 5: PSL benchmark matrix (proposal Table 1). Best: Transformer + Full FT = 30.00%.

Team Contributions

All team members contributed to project planning, experiment design, and review across every stage of the work.

Sameeha Shahid — Led ASL source validation for both I3D and Pose-TGCN on WLASL-100. Set up the pretrained checkpoints, evaluation pipeline, and baseline comparison plots.

Emaim Mohsin — Ran the PSL transfer experiments: TGCN and Pose Transformer four-stage benchmarks on WLPSSL, including confusion matrices and result packaging.

Fatima Amin — Handled WLPSSL dataset preparation and initial validation: video organization, pose extraction, and early PSL experiment setup.

Mishaal Mustazhar — Built the ASL→FSL transfer benchmark: LSFBS-ISOL poses, Pose Transformer pretraining, and the full four-strategy result matrix.

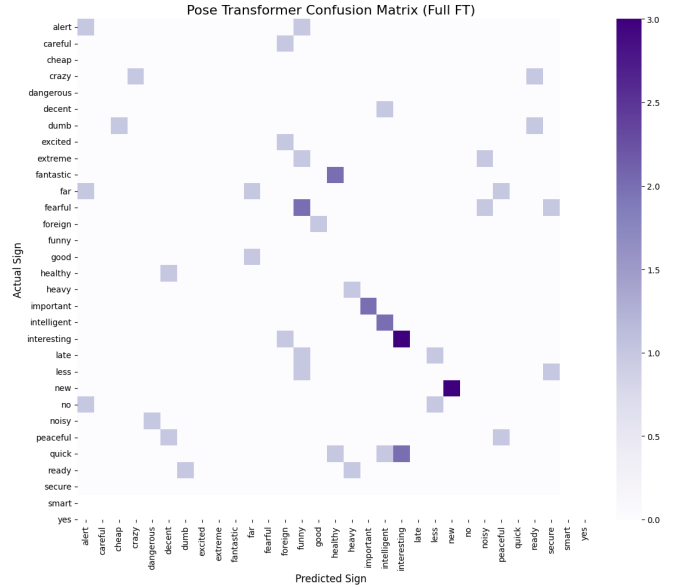


Figure 6: PSL Pose Transformer confusion matrix (30%).

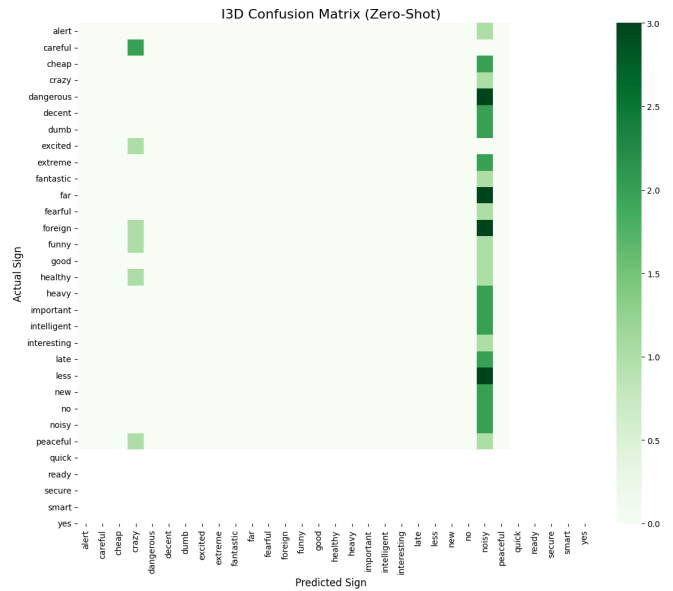


Figure 7: PSL I3D confusion matrix.